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Date: 3/2/26 **Class:** 1

MyDesign Portfolio

Element A

Mr. Connolly and the film festival students regularly run into problems when recording audio with boom microphones during student film projects. Boom microphones are meant to capture clear dialogue without being seen on camera, but the attached cables often hang down into the shot or get in the way while filming. This issue happens in classrooms, auditoriums, and other school filming locations where students record scenes for projects and festivals. Because students are usually working in tight spaces with limited equipment, keeping microphone cables organized becomes a constant challenge



(Figure 1) Cable wrapped around the pole

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(Figure 2) Cable hang from the pole

The problem is when recording, the wires are getting in the way when using the boom mics. This is important because Mr. Connolly and the film students need to have clear shots for their movie without cable interference. A current available solution is wrapping the cable around the boom, which damages the cables. Mr. Connolly and the film students need a solution to hold the cable to the boom mic and keep them straight. So in order to relieve this problem our team will design a way for the wires to be attached to the boom that is inexpensive, light weight, durable & adjustable with quantifiable constraints such as a unit cost of about \$2 and a weight of approximately 3 grams.

This problem happens every time filming takes place with boom microphones, especially during rehearsals and scenes that require multiple takes. Since there is no built-in way to secure the cable to the boom pole, students often wrap the cable around the pole (Figure 2) to keep it out of frame. While this may seem like a quick fix, repeatedly bending and twisting cables can damage the wires inside and shorten their lifespan, according to the Audio Engineering Society (2018). Sweetwater, a professional audio equipment retailer, also explains that poor cable management is one of the most common causes of audio signal failure and cable replacement in recording environments (Sweetwater, 2024).

The issue goes beyond just one class or project. The U.S. Bureau of Labor Statistics reports that damage to audio and video equipment is a recurring cost in educational media programs, especially when equipment is shared by many students (2022). In addition, the National



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Association of Broadcasters states that cable failures are a leading cause of audio problems during recording and live production, often leading to lost time and repeated takes (2019). Because boom microphones are used in nearly every dialogue-heavy scene, cable interference can happen multiple times during a single filming session.

This problem exists largely because boom microphone systems are typically sold without built-in cable attachment solutions, assuming users will manage cables separately. In school environments, students may not have access to professional accessories or extensive training, which makes improper cable handling more likely. Research published by the Audio Engineering Society shows that unmanaged cables experience more stress and fail more often in places where equipment is frequently set up and taken down, such as schools (2018).

Several primary stakeholders are directly affected by this problem. Film students are impacted because cables appearing in shots can ruin footage and force retakes, which takes away from learning time. Mr. Connolly, as the teacher and equipment manager, must deal with damaged cables that require repair or replacement. School media programs are also affected because frequent equipment damage increases costs and limits available resources. Future students are impacted as well, since damaged equipment may not be available or functional for later classes. Finally, student film crews are affected because cable issues slow down production and make collaboration more difficult. Secondary stakeholders include school administrators who must fund replacements and the school community members that view student films at events and festivals.

This problem needs to be addressed now because continued cable damage leads to higher costs, wasted time, and lower-quality student work. Sweetwater reports that professional XLR microphone cables typically cost between \$15 and \$40 each, which can quickly become expensive for schools if replacements are frequent (Sweetwater, 2024). The National Education Association emphasizes that reducing avoidable equipment costs allows schools to focus funding on instruction and learning resources (2021). Our team will create an inexpensive, lightweight, durable, and adjustable boom mic cable attachment—costing about \$2 and weighing roughly 3 grams—that will help protect equipment, improve filming efficiency, and support long-term sustainability in school media programs.

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Name: Richard B, Niko M, Seth N **Date:** 3/2/26 **Class:** 1

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Figure 1, Bock, R. Cable Wrapped. 11 Feb, 2026. Author's Personal Collection.

Figure 2, Bock, R. Cable Hanging. 11 Feb, 2026. Author's Personal Collection.

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